

Recitation 1: Introduction to Stata and Hypothesis Testing

ECON UN3412: Econometrics

Faiz Essa

September 13, 2026

1. Introduction to Econometrics
2. Stata Basics & Workflow
3. Loading and Saving Data
4. Inspecting & Modifying Data
5. Two-Sided Hypothesis Testing in Stata

1. Introduction to Econometrics

What is Econometrics?

- Application of statistical methods to economic data.

What is Econometrics?

- Application of statistical methods to economic data.
 - Estimating parameters in economic models

What is Econometrics?

- Application of statistical methods to economic data.
 - Estimating parameters in economic models
 - Using historical patterns to *predict* future economic outcomes

What is Econometrics?

- Application of statistical methods to economic data.
 - Estimating parameters in economic models
 - Using historical patterns to *predict* future economic outcomes
 - Most importantly, inference on the relationship between economic variables, especially **causal** relationships

What is Econometrics?

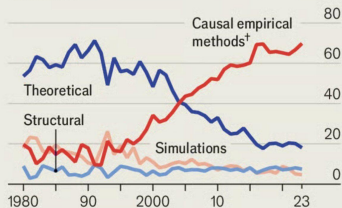
- Application of statistical methods to economic data.
 - Estimating parameters in economic models
 - Using historical patterns to *predict* future economic outcomes
 - Most importantly, inference on the relationship between economic variables, especially **causal** relationships
- These skills are extremely valuable to any quantitative field.

History of Data in Economics

1. Qualitative & Philosophical (Pre-20th C.)

Data deluge

NBER and CEPR working papers*, % of total
By method



*44,800 papers published by National Bureau of Economic Research and Centre for Economic Policy Research

†Includes instrumental variables, randomised controlled trials, etc

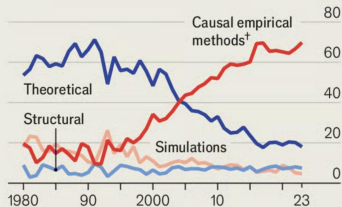
Source: "Causal claims in economics",
by P. Garg and T. Fetzer, 2025 (pre-print)

CHART: THE ECONOMIST

History of Data in Economics

Data deluge

NBER and CEPR working papers*, % of total
By method



*44,800 papers published by National Bureau of Economic Research and Centre for Economic Policy Research

†Includes instrumental variables, randomised controlled trials, etc
Source: "Causal claims in economics",
by P. Garg and T. Fetzer, 2025 (pre-print)

CHART: THE ECONOMIST

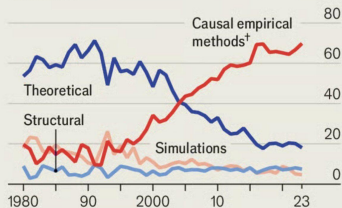
1. Qualitative & Philosophical (Pre-20th C.)

- Descriptive political economy and deduction (Smith, Ricardo).

History of Data in Economics

Data deluge

NBER and CEPR working papers*, % of total
By method



*44,800 papers published by National Bureau of Economic Research and Centre for Economic Policy Research

†Includes instrumental variables, randomised controlled trials, etc
Source: "Causal claims in economics",
by P. Garg and T. Fetzer, 2025 (pre-print)

CHART: THE ECONOMIST

1. Qualitative & Philosophical (Pre-20th C.)

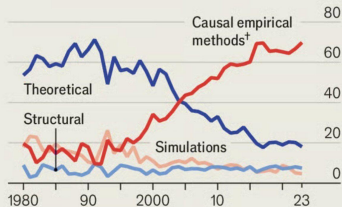
- Descriptive political economy and deduction (Smith, Ricardo).

2. Pure Theory (Early–Mid 20th C.)

History of Data in Economics

Data deluge

NBER and CEPR working papers*, % of total
By method



*44,800 papers published by National Bureau of Economic Research and Centre for Economic Policy Research

†Includes instrumental variables, randomised controlled trials, etc
Source: "Causal claims in economics",
by P. Garg and T. Fetzer, 2025 (pre-print)

CHART: THE ECONOMIST

1. Qualitative & Philosophical (Pre-20th C.)

- Descriptive political economy and deduction (Smith, Ricardo).

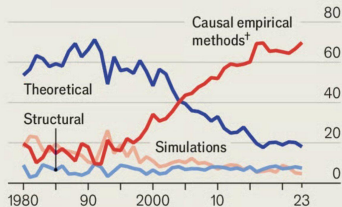
2. Pure Theory (Early–Mid 20th C.)

- Formal mathematical models and axiomatic proofs (Walras, Arrow-Debreu).

History of Data in Economics

Data deluge

NBER and CEPR working papers*, % of total
By method



*44,800 papers published by National Bureau of Economic Research and Centre for Economic Policy Research

†Includes instrumental variables, randomised controlled trials, etc
Source: "Causal claims in economics",
by P. Garg and T. Fetzer, 2025 (pre-print)

CHART: THE ECONOMIST

1. Qualitative & Philosophical (Pre-20th C.)

- Descriptive political economy and deduction (Smith, Ricardo).

2. Pure Theory (Early–Mid 20th C.)

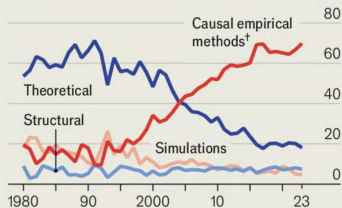
- Formal mathematical models and axiomatic proofs (Walras, Arrow-Debreu).

3. The Empirical Turn (Late 20th C.)

History of Data in Economics

Data deluge

NBER and CEPR working papers*, % of total
By method



*44,800 papers published by National Bureau of Economic Research and Centre for Economic Policy Research

[†]Includes instrumental variables, randomised controlled trials, etc
Source: "Causal claims in economics",
by P. Garg and T. Fetzer, 2025 (pre-print)

CHART: THE ECONOMIST

1. Qualitative & Philosophical (Pre-20th C.)

- Descriptive political economy and deduction (Smith, Ricardo).

2. Pure Theory (Early–Mid 20th C.)

- Formal mathematical models and axiomatic proofs (Walras, Arrow-Debreu).

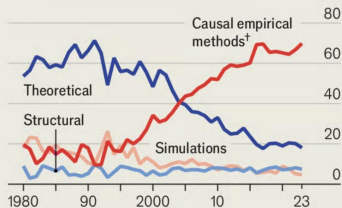
3. The Empirical Turn (Late 20th C.)

- Rise of computing and regression analysis on observational datasets.

History of Data in Economics

Data deluge

NBER and CEPR working papers*, % of total
By method



*44,800 papers published by National Bureau of Economic Research and Centre for Economic Policy Research

[†]Includes instrumental variables, randomised controlled trials, etc
Source: "Causal claims in economics",
by P. Garg and T. Fetzer, 2025 (pre-print)

CHART: THE ECONOMIST

1. Qualitative & Philosophical (Pre-20th C.)

- Descriptive political economy and deduction (Smith, Ricardo).

2. Pure Theory (Early–Mid 20th C.)

- Formal mathematical models and axiomatic proofs (Walras, Arrow-Debreu).

3. The Empirical Turn (Late 20th C.)

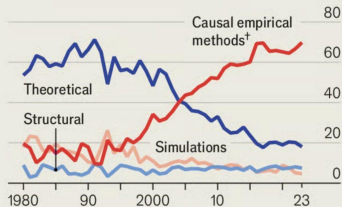
- Rise of computing and regression analysis on observational datasets.

4. Causal Empirical Work (1990s–2000s)

History of Data in Economics

Data deluge

NBER and CEPR working papers*, % of total
By method



*44,800 papers published by National Bureau of Economic Research and Centre for Economic Policy Research

[†]Includes instrumental variables, randomised controlled trials, etc
Source: "Causal claims in economics",
by P. Garg and T. Fetzer, 2025 (pre-print)

CHART: THE ECONOMIST

1. Qualitative & Philosophical (Pre-20th C.)

- Descriptive political economy and deduction (Smith, Ricardo).

2. Pure Theory (Early–Mid 20th C.)

- Formal mathematical models and axiomatic proofs (Walras, Arrow-Debreu).

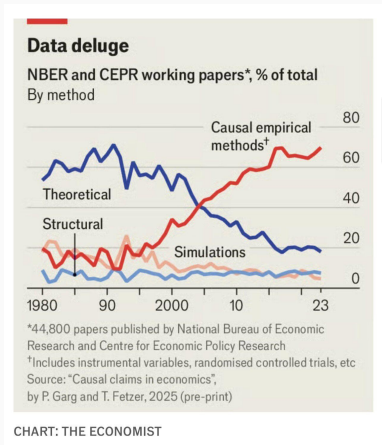
3. The Empirical Turn (Late 20th C.)

- Rise of computing and regression analysis on observational datasets.

4. Causal Empirical Work (1990s–2000s)

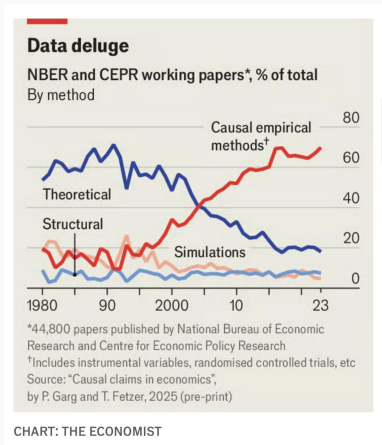
- The *Credibility Revolution*: randomized control trials and “natural” experiments to uncover causal effects.

History of Data in Economics



1. **Qualitative & Philosophical** (Pre-20th C.)
 - Descriptive political economy and deduction (Smith, Ricardo).
2. **Pure Theory** (Early–Mid 20th C.)
 - Formal mathematical models and axiomatic proofs (Walras, Arrow-Debreu).
3. **The Empirical Turn** (Late 20th C.)
 - Rise of computing and regression analysis on observational datasets.
4. **Causal Empirical Work** (1990s–2000s)
 - The *Credibility Revolution*: randomized control trials and “natural” experiments to uncover **causal** effects.
5. **Replicable Science** (Today)

History of Data in Economics



1. Qualitative & Philosophical (Pre-20th C.)

- Descriptive political economy and deduction (Smith, Ricardo).

2. Pure Theory (Early–Mid 20th C.)

- Formal mathematical models and axiomatic proofs (Walras, Arrow-Debreu).

3. The Empirical Turn (Late 20th C.)

- Rise of computing and regression analysis on observational datasets.

4. Causal Empirical Work (1990s–2000s)

- The *Credibility Revolution*: randomized control trials and “natural” experiments to uncover causal effects.

5. Replicable Science (Today)

- Open data and reproducible code pipelines.

Why Stata?

- Computation and more data have allowed economics to become an empirical field.
- Stata is a widely-used statistical package in economics and social sciences.
- Other options: R, Python, Julia.
- Stata is user-friendly and doesn't require a traditional coding background.

Accessing Stata at Columbia

- **Download through Columbia (Recommended):**
 - Download via CUIT: cuit.columbia.edu/content/stata (login with your UNI).
 - The latest version available is **Stata 19**.
- **On-Campus Computer Labs:**
 - Find lab locations & hours: cuit.columbia.edu/.../locations-hours.
- **Remote PC Access:**
 - Access Stata remotely from your PC: cuit.columbia.edu/computer-lab-technologies.

Important Caveats:

- Not all systems have the current Stata version (won't affect this course).
- Remote availability can be limited during peak times — **start problem sets early!**

2. Stata Basics & Workflow

- **Command Window:** For typing quick, interactive commands.
- **Results Window:** Displays command history and formatted numerical output.
- **Variables & Properties:** Shows variable names, storage types, formats, and labels.
- **Review Window:** Lists previously executed commands.
- **Always work from a do-file (.do),** *not* from the interactive console. This guarantees that your research is 100% **reproducible** and auditable.

Setting Up a Clean Do-File Header

Start every do-file with standard initialization commands:

```
clear all           // clear existing data from memory
capture log close   // close any open log files
set more off        // disable pause prompts during long output
log using "recitation1.log", replace text
```

When your script finishes, close the log:

```
log close
```

Your do-file should import your data and produce the same results every time. I highly recommend you do this on your problem sets.

- The `.log` file saves the output of your do-file.
- In practice economists often instead produce tables of their results using `esttab` or `estout`.
- For this class, simply screenshotting results directly from the console or a log-file will be sufficient.

3. Loading and Saving Data

- Stata includes pre-installed datasets for practice.
- We will use the 1978 Automobile dataset (`auto.dta`):

```
sysuse auto, clear
```

- The `clear` option discards whatever is in memory so the new dataset loads cleanly.

Saving and Loading Data: Stata Format (.dta)

- Set working directory:

```
cd "/Users/faizessa/Library/CloudStorage/Dropbox/Mac/Documents/ta/ECONU"
```

- Saving data to Stata format (.dta):

```
save "auto.dta", replace
```

- Importing data from Stata format (.dta):

```
clear
```

```
use "auto.dta"
```

Saving and Loading Data: CSV Format (.csv)

- **Exporting data to CSV format (.csv):**

```
export delimited using "auto.csv", replace
```

- **Importing data from CSV format (.csv):**

```
clear  
import delimited "auto.csv"
```

Saving and Loading Data: Excel Format (.xlsx)

- Exporting data to Excel format (.xlsx):

```
export excel using "auto.xlsx", firstrow(variables) replace
```

- Importing data from Excel format (.xlsx):

```
clear  
import excel "auto.xlsx", firstrow
```

(Note: firstrow tells Stata that row 1 contains the variable names)

4. Inspecting & Modifying Data

Inspecting the Dataset

```
summarize mpg price weight
```

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
mpg	74	21.2973	5.785503	12	41
price	74	6165.257	2949.496	3291	15906
weight	74	3019.459	777.1936	1760	4840

- describe: Displays storage formats, variable labels, and value labels.
- summarize: Computes observations, sample mean, standard deviation, min, and max.
- browse: Opens a read-only spreadsheet viewer for inspection.

The edit Function vs. Scripted Editing

- In Stata, the `edit` command opens the interactive Data Editor GUI:

```
edit          // opens editable spreadsheet window
```

- **Avoid Manual edit:**
 - Manual point-and-click changes cannot be tracked, audited, or replicated.
 - One accidental keystroke can corrupt your dataset.
- *Always* use Stata commands (`generate`, `replace`) in your do-file so every change is recorded.

Scripted Transformations: generate & replace

```
// Create gallons per 100 miles
generate gpm = (1 / mpg) * 100

// Create a dummy indicator for high fuel economy (mpg >= 25)
generate high_mpg = (mpg >= 25)

summarize gpm high_mpg
```

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
gpm	74	5.01928	1.279856	2.439024	8.333333
high_mpg	74	.2567568	.439826	0	1

- Use generate to create a brand new variable.
- Use replace to update values of an existing variable.

5. Two-Sided Hypothesis Testing in Stata

Setting Up the Two-Sided Hypothesis

- **Research Question:** Is the average fuel economy of 1978 cars different from 20 mpg?
- **Parameter of Interest:** Population mean $\mu = E[\text{mpg}]$
- **Two-Sided Hypotheses:**

$$H_0 : \mu = 20 \quad \text{vs.} \quad H_1 : \mu \neq 20$$

- **Significance Level:** $\alpha = 0.05$ ($\alpha/2 = 0.025$ in each tail).

Step 1: Summary Statistics from Stata

```
summarize mpg
```

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
mpg	74	21.2973	5.785503	12	41

From the summary table, we extract our sample estimators:

- Sample size: $n = 74$
- Sample mean: $\bar{Y} = 21.2973$
- Sample standard deviation: $S = 5.7855$ ($S^2 = 33.472$)

Step 2: Critical Value & p -Value Approaches (by Hand)

1. Standard Error & t -Statistic:

$$SE(\bar{Y}) = \frac{5.7855}{\sqrt{74}} = 0.67255, \quad t^{\text{act}} = \frac{21.2973 - 20}{0.67255} = 1.9289$$

Step 2: Critical Value & p -Value Approaches (by Hand)

1. Standard Error & t -Statistic:

$$SE(\bar{Y}) = \frac{5.7855}{\sqrt{74}} = 0.67255, \quad t^{\text{act}} = \frac{21.2973 - 20}{0.67255} = 1.9289$$

2. Method 1: Critical Value Rule ($\alpha = 0.05$):

- Two-sided threshold: $c_{\alpha/2} = 1.960$.
- Since $|t^{\text{act}}| = 1.9289 \leq 1.960 \implies$ **Fail to reject H_0 at 5%.**

Step 2: Critical Value & p -Value Approaches (by Hand)

1. Standard Error & t -Statistic:

$$SE(\bar{Y}) = \frac{5.7855}{\sqrt{74}} = 0.67255, \quad t^{\text{act}} = \frac{21.2973 - 20}{0.67255} = 1.9289$$

2. Method 1: Critical Value Rule ($\alpha = 0.05$):

- Two-sided threshold: $c_{\alpha/2} = 1.960$.
- Since $|t^{\text{act}}| = 1.9289 \leq 1.960 \implies$ **Fail to reject H_0 at 5%.**

3. Method 2: p -Value Rule ($\alpha = 0.05$):

- $p\text{-value} = 2 \cdot [1 - \Phi(1.9289)] = 0.0576$.
- Since $p\text{-value} = 0.0576 > 0.05 \implies$ **Fail to reject H_0 at 5%.**

Step 3: Confidence Interval Approach (by Hand)

Method 3: Construct the 95% Confidence Interval:

$$CI_{95\%} = [\bar{Y} - 1.960 \cdot SE(\bar{Y}), \bar{Y} + 1.960 \cdot SE(\bar{Y})]$$

$$CI_{95\%} = [21.2973 - 1.960(0.67255), 21.2973 + 1.960(0.67255)] = [19.979, 22.616]$$

- **Duality Rule:** Reject $H_0 : \mu = \mu_0$ at level $\alpha \iff \mu_0 \notin CI_{1-\alpha}$.
- **Decision:** Since $\mu_0 = 20 \in [19.979, 22.616]$, we **fail to reject H_0 at the 5% level**.

***Equivalence:** Critical Value, p-Value, and Confidence Interval approaches are mathematically identical and always yield the same decision.*

Step 4: Automated Testing with `ttest`

In practice, Stata computes all three approaches simultaneously with a single command:

```
ttest mpg == 20
```

One-sample t test

Variable	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
mpg	74	21.2973	.6725511	5.785503	19.9569	22.63769

mean = mean(mpg) t = 1.9289
Ho: mean = 20 degrees of freedom = 73

Ha: mean < 20	Ha: mean != 20	Ha: mean > 20
Pr(T < t) = 0.9712	Pr(T > t) = 0.0576	Pr(T > t) = 0.0288

Comparing ttest Output to Our 3 Approaches

Stata's ttest output displays all three methods in a single table:

- **1. Critical Value:** $t = 1.9289$ (with $df = 73$) compared against ± 1.960 .
- **2. p -Value:** Center column gives two-sided p -value = 0.0576 (> 0.05).
- **3. Confidence Interval:** Right columns report [19.9569, 22.6377] (contains 20).
- **Universal Conclusion at $\alpha = 0.05$:** **Fail to reject** $H_0 : \mu = 20$ in favor of $H_1 : \mu \neq 20$.

Stata's automated output identically mirrors our manual calculations across all 3 testing approaches.

Useful Stata Commands & Resources

Commands Covered Today:

- **Load/Save:** sysuse, use, save, import, export
- **Inspect:** describe, summarize (detail)
- **Variables:** generate, replace
- **Inference:** ttest

Coming Up in Econometrics:

- **Regression:** regress (OLS estimation)
- **Association:** correlate (correlation & covariance)
- **Advanced Variables:** egen (group stats, rankings)
- **Merging:** merge, append
- **Visuals:** twoway scatter, lfit, graph

Recommended Online Resources:

- UCLA IDRE Stata Portal
(Tutorials, code examples, & modules)
- Official Stata Cheat Sheets
(Syntax & command quick-reference)
- Statalist Discussion Forum
(Q&A community for econometrics)

Built-in Stata Help:

Type `help <command>` (e.g., `help regress`) for documentation, syntax options, and return values.